

Table of Contents

● Overall Architecture Flow	-----	3
● Navigation Structure (Sidebar Menu)	-----	5
● Dashboard Flow	-----	7
● Users Management Flow	-----	9
● Trades & Orders Flow	-----	11
● Portfolio & P&L Flow	-----	13
● Market Data Flow	-----	15
● Notifications Flow	-----	17
● Platform Settings Flow	-----	19
● Security & Audit Flow	-----	21
● Real-time Update Flow	-----	22

1. Overall Architecture Flow

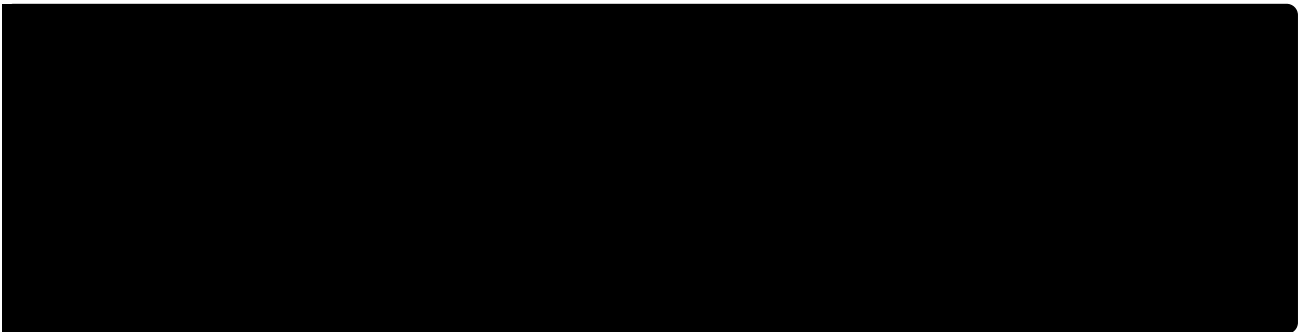
The Pepertect Admin Panel follows a modern full-stack architecture with real-time capabilities. The admin interface communicates with backend APIs that interact with the database and broadcast updates via WebSocket connections to connected client applications.

System Architecture Overview



Connection Details

Real-time Data Flow Architecture



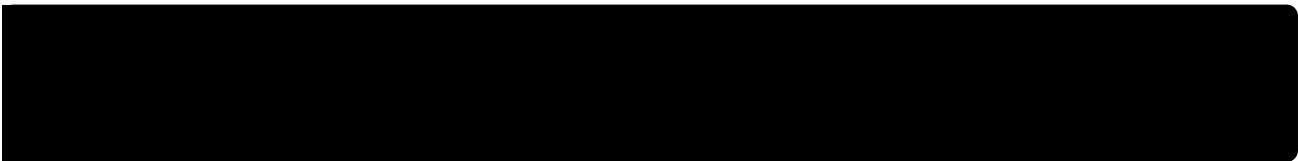
External Integrations

Upstox API
Market data provider. WebSocket connection for live stock prices. REST API for historical data. Token-based authentication with auto-refresh.

Socket.io Server
Runs as a separate mini-service on port 3003. Handles all real-time communication between admin panel and user frontend. Supports rooms for targeted broadcasts.

Prisma ORM
Database abstraction layer connecting to SQLite. Provides type-safe queries, migrations, and schema management. Client imported from @///lib/db.

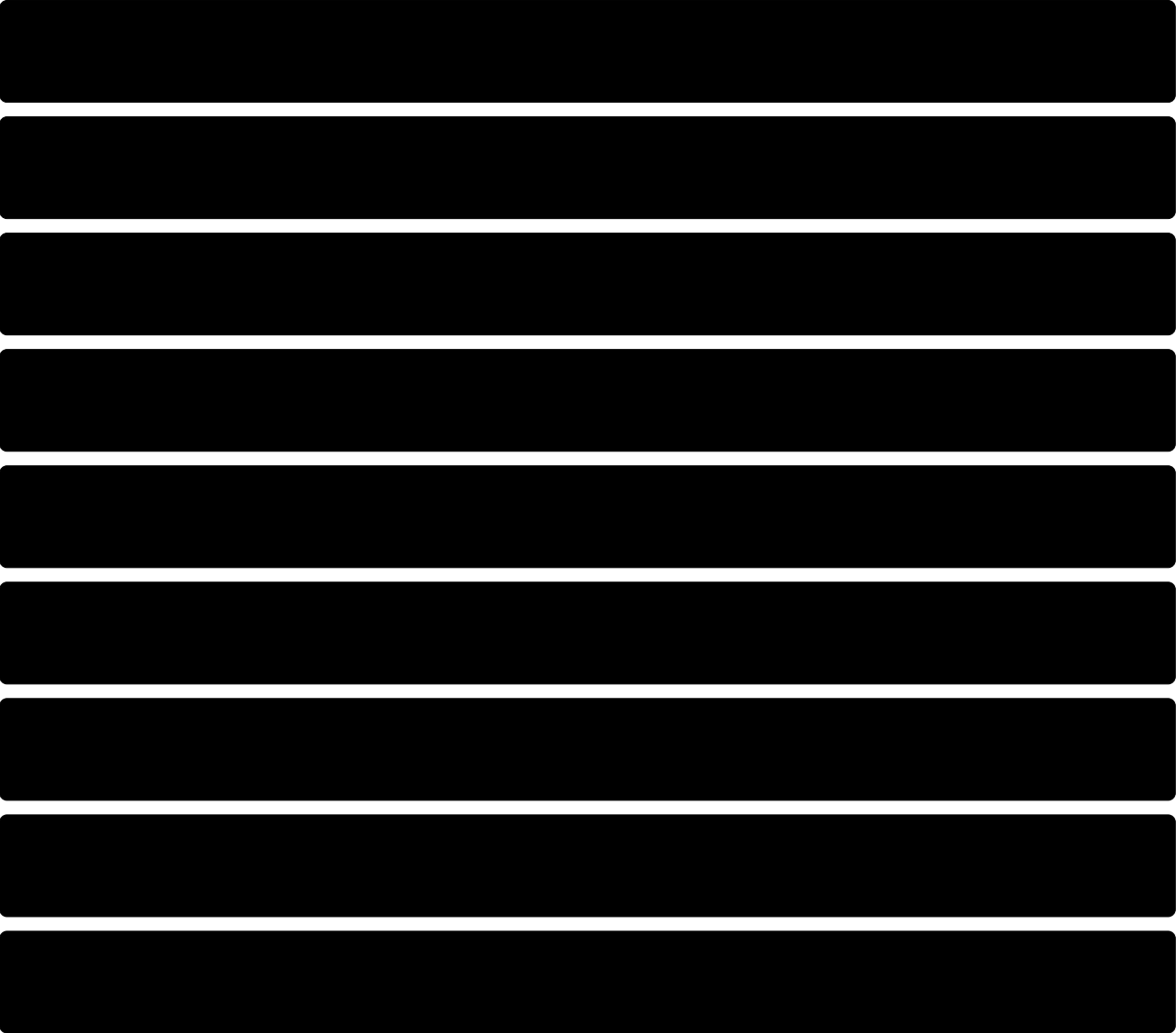
Authentication & Authorization Flow



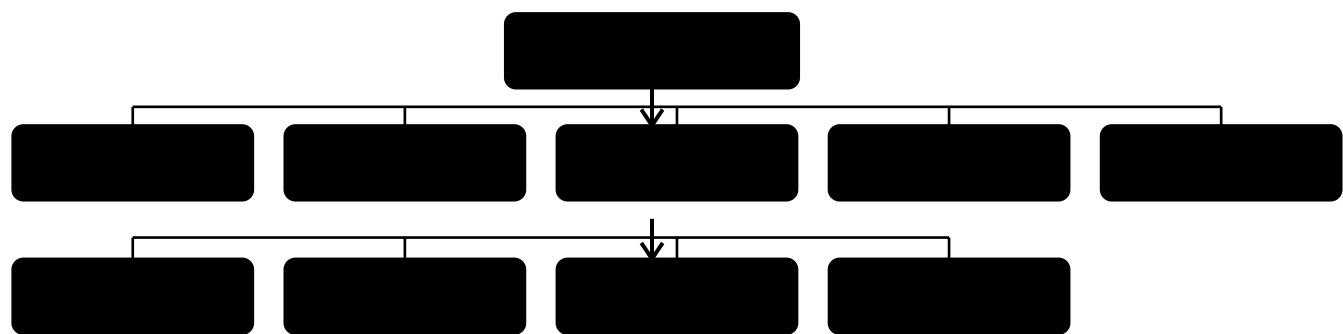
2. Navigation Structure (Sidebar Menu)

The admin panel uses a persistent sidebar navigation that remains fixed on the left side of the screen. Each menu item provides access to a specific functional area of the platform.

Sidebar Menu Items



Navigation Flow Diagram



Navigation Behavior

- Sidebar is always visible on desktop (width "e 1024px), collapsible on mobile
- Active menu item is highlighted with green (#00D09C) background
- Breadcrumb navigation shows current location: Admin > Section > Page
- Each section has a list view (default) and detail view (on item click)
- Quick search accessible via Ctrl+K shortcut for fast navigation
- Notification badge count updates in real-time via WebSocket

3. Dashboard Flow

Dashboard Access Flow



Stats Cards Section

Total Users —	Active Traders —	Today's Trades —	Platform Volume —
------------------	---------------------	---------------------	----------------------

Charts Section

- User Growth Chart — Line chart showing daily/weekly/monthly user registrations
- Trade Volume Chart — Bar chart displaying trade volume over time periods
- Active Users Heatmap — Visual representation of peak usage hours
- Revenue Distribution — Pie chart of virtual trading volume by segment

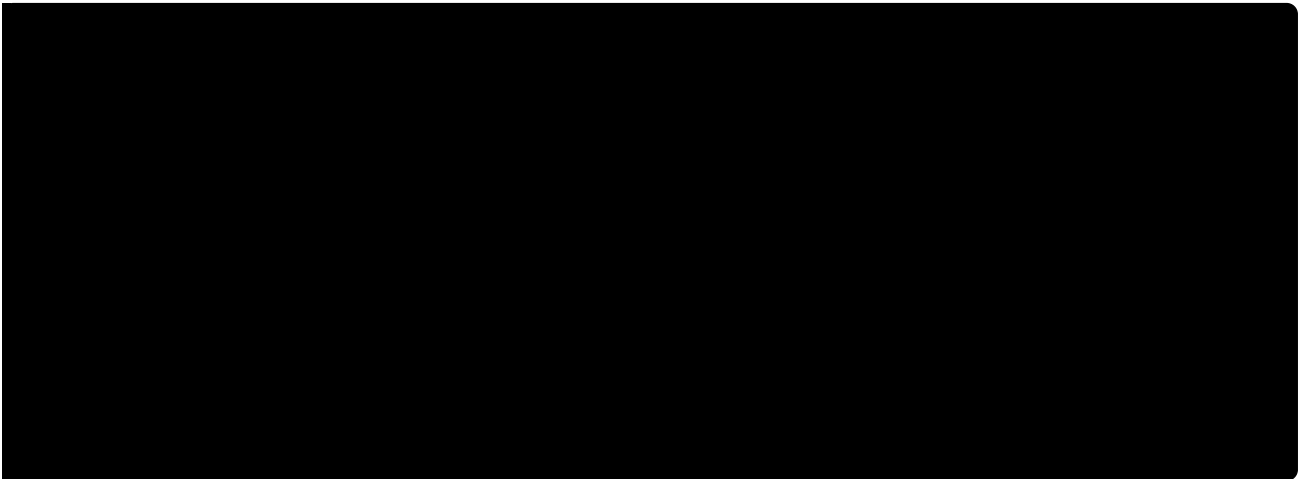
System Health Monitor

Upstox WS Status Connected	API Response Time < 200ms	Server Uptime 99.9%
-------------------------------	------------------------------	------------------------

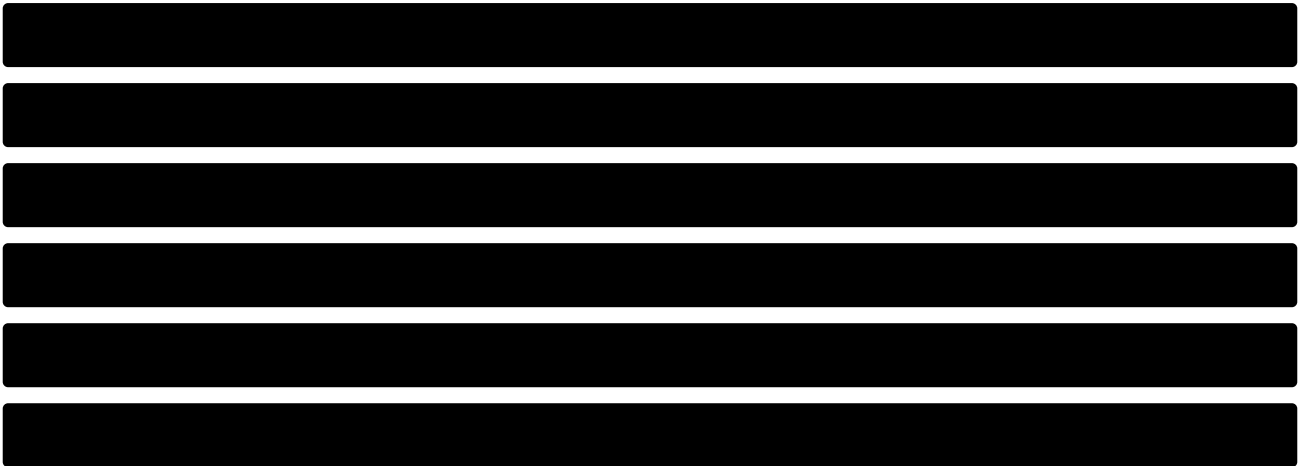
Quick Actions



Dashboard Data Flow



Dashboard API Endpoints



4. Users Management Flow

Users Management Flow Diagram



Users List Page

- Search Bar — Full-text search across name, email, and phone number fields
- Filters — Role (User/Admin), Status (Active/Banned/Suspended), Registration date range
- Sort Options — Name, Join date, Last active, Trade count, Portfolio value
- Pagination — 20 users per page with infinite scroll option
- Bulk Actions — Select multiple users !' Ban, Export, Send notification

User Detail Page



User Actions Detail

Edit User

Modify user profile fields: name, email, phone, virtual cash balance. Changes are logged in audit trail.

Reset Password

Generate a temporary password and send via email. Forces password change on next login.

Ban/Unban User

Toggle user account status. Banned users cannot log in. Active sessions are immediately invalidated via Socket.io.

Delete User

Permanently remove user and all associated data (trades, portfolio, notifications). This action is IRREVERSIBLE.

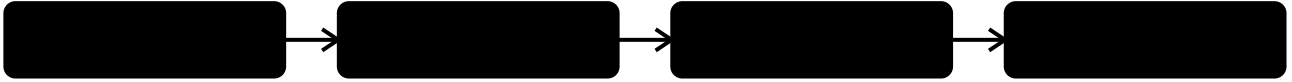
Promote to Admin

Elevate user role to admin with full access. Requires super-admin privilege. Logged in audit trail.

5. Real-time notification sent to affected user (if applicable)

5. Trades & Orders Flow

Trades & Orders Flow Diagram



Trades List Page

Displays all trades across the platform with comprehensive filtering and sorting capabilities.

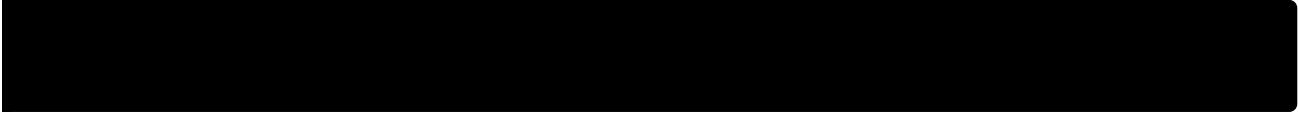
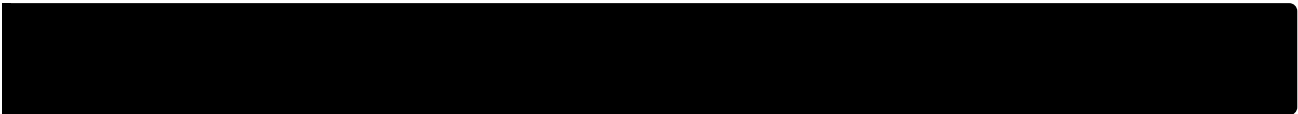
- User filter — Search by username or user ID
- Stock filter — Search by stock symbol or company name
- Date range — From/To date picker for trade execution time
- Trade type — BUY / SELL / ALL
- Order type — MARKET / LIMIT / STOP LOSS
- Status — COMPLETED / PENDING / CANCELLED / REJECTED

Open Positions View

Real-time view of all current holdings across users. Data updates via WebSocket connection to ensure positions reflect live market prices.

- User name and ID
- Stock symbol and company name
- Quantity held
- Average entry price
- Current market price (live)
- Unrealized P&L (absolute and percentage)
- Day change (absolute and percentage)

Order Management



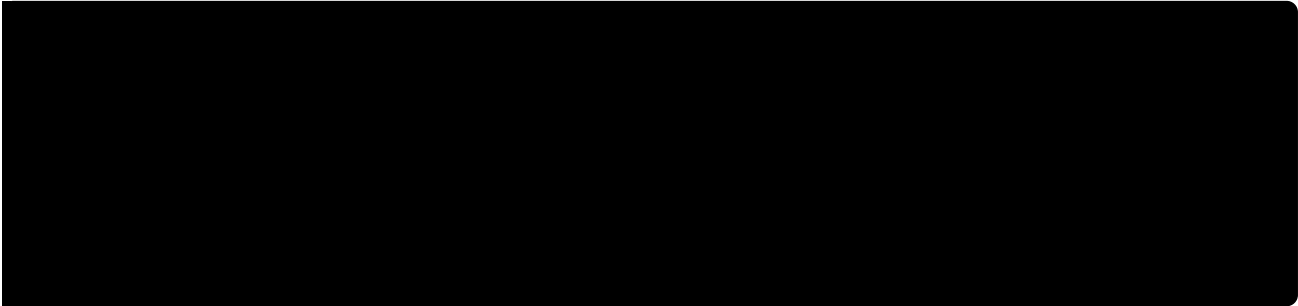
Trade Statistics

Total Volume
Aggregate number of shares traded today

Popular Stocks
Top 10 most traded stocks by volume

Active Hours
Heatmap of trading activity by hour

Buy vs Sell Ratio
Ratio of buy orders to sell orders



Socket event: "trade-update" ! { type, tradeid, userid, data }

6. Portfolio & P&L Flow

Portfolio Flow Diagram



All Portfolios Summary

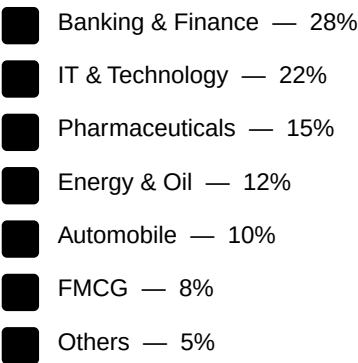
A comprehensive table showing all user portfolios with key metrics. Supports sorting, searching, and filtering.

Top Gainers & Losers

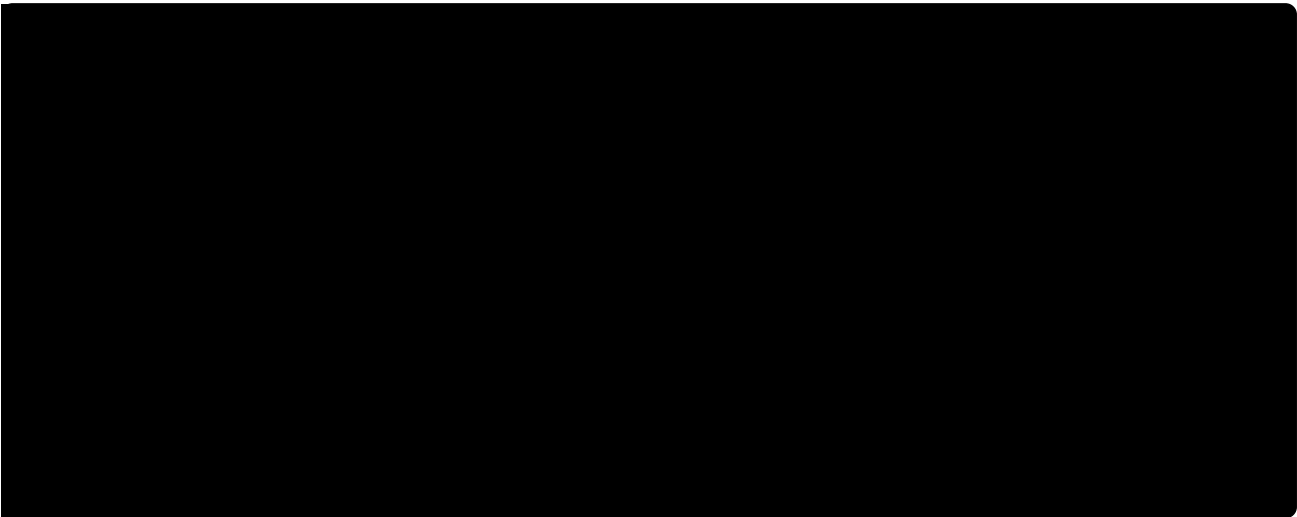
1. Amit K.	+12.0%	1. Vikram P.	-15.2%
2. Rahul S.	+9.0%	2. Priya M.	-8.3%
3. Sara J.	+5.0%	3. Neha R.	-6.1%

Sector-wise Distribution

Pie chart showing portfolio allocation across market sectors:



Balance Overview

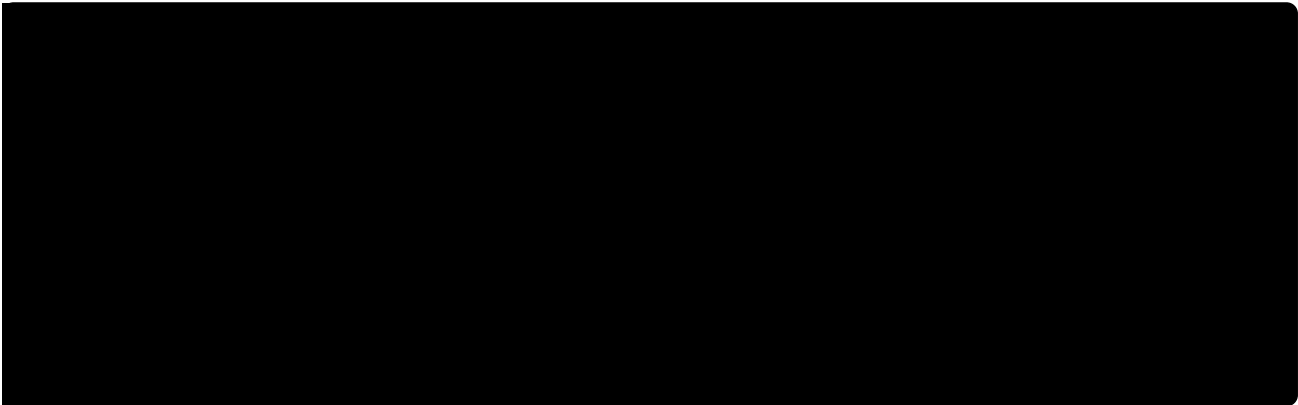


7. Market Data Flow

Market Data Flow Diagram



Upstox Connection Status



WebSocket Feed Status

Feed Connection		WebSocket connection to Upstox feed is active
Price Updates		Live price updates are being received for subscribed stocks
Depth Data		Market depth data (Level 2) is available
Index Data		NIFTY 50 and SENSEX data is being received

API Usage Monitoring

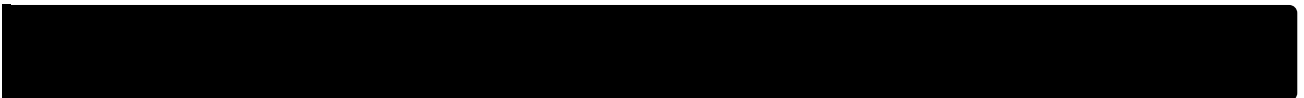
Track Upstox API usage to avoid rate limiting and ensure optimal performance:

- Requests per minute (current vs limit)
- Total requests today
- Error rate (4xx / 5xx responses)
- Average response time
- Rate limit warnings count
- Auto-throttling events

Cache Management

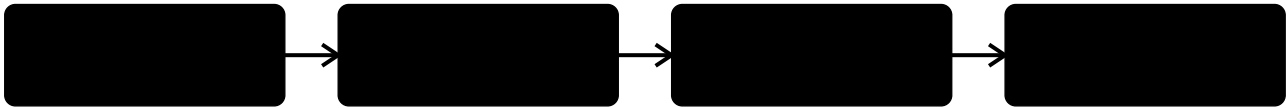


Stock List Management



8. Notifications Flow

Notification Flow Diagram



Notification Types

Ø=Üâ Broadcast to All Users

Send a notification to every registered user. Appears in the notification bell icon and as an in-app toast. Can optionally trigger push notification (if PWA enabled).

' p Individual User Notification

Send a targeted notification to a specific user. Useful for account-related messages, warnings, or personalized updates.

#ð Scheduled Notification

Create a notification to be sent at a future date/time. Supports cron expressions for recurring notifications. Can be cancelled before delivery.

Ø=þ© Announcement Banner

Display a persistent banner at the top of the user interface. Remains visible until dismissed by user or removed by admin. Supports HTML formatting.

Notification Creation Form

Notification History & Delivery Tracking

Socket event: "user-notification:[userId]" : { title, message }

9. Platform Settings Flow

Settings Categories

Ø=Ý Upstox Token Management

View current token status, manually refresh tokens, set up auto-refresh intervals, view token history and expiry logs.

Ø=Üµ Virtual Cash Configuration

Set default virtual cash for new users (currently 1,00,000), configure min/max balance limits, set up periodic balance resets.

Ø=ÝP Trading Hours Setup

Configure market hours (9:15 AM - 3:30 PM IST), pre-market and post-market sessions, holiday calendar management.

Ø=Ý" Maintenance Mode

Toggle maintenance mode on/off. When enabled, users see a maintenance banner and cannot place trades. Admin can set a custom message and estimated recovery time.

Ø=P© Feature Flags

Enable/disable platform features independently: F&O trading, Options trading, Short selling, After-market orders, Social features (leaderboard, following).

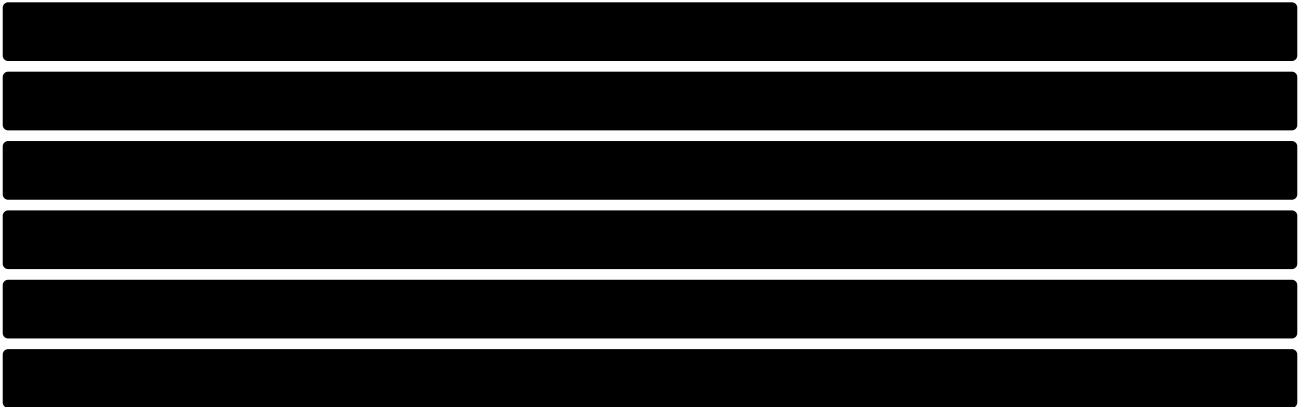
&j Rate Limit Configuration

Set API rate limits per user tier: Default (60 req/min), Premium (120 req/min), Admin (unlimited). Configure cooldown periods and auto-ban thresholds.

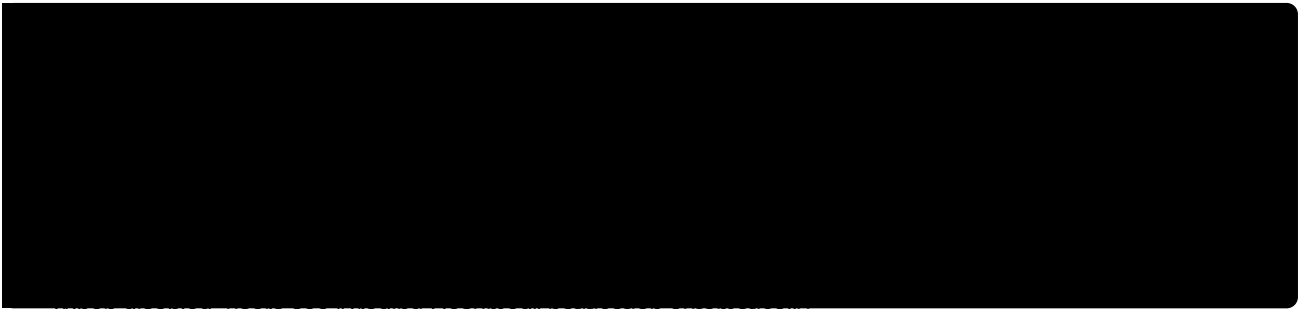
Upstox Token Management Detail



Feature Flags Configuration



Maintenance Mode Flow



- When disabled, users see "Trading Resumed" notification automatically

10. Security & Audit Flow

Security Monitoring Overview

Ø=ÜË Admin Action Logs

Complete audit trail of all admin actions. Every API call is logged with: admin ID, action type, target resource, timestamp, IP address, request body (sanitized). Logs are immutable and retained for 90 days.

Ø=Ý Suspicious Activity Detection

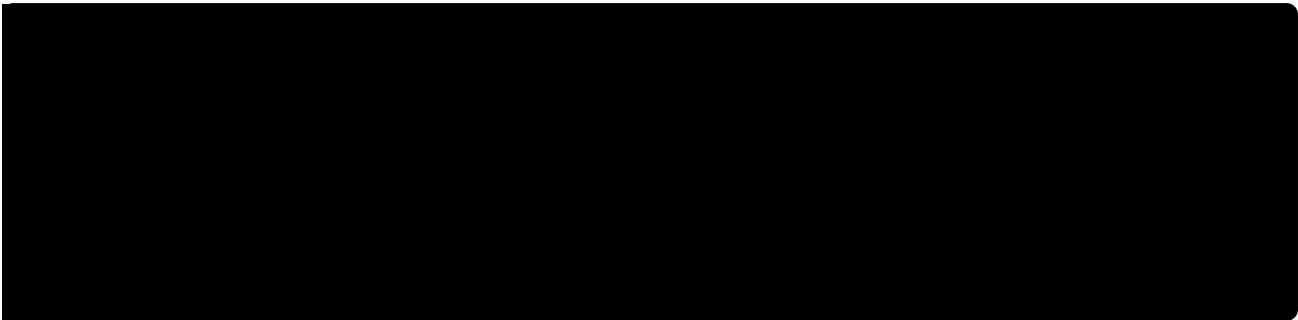
Automated detection of unusual patterns: Multiple failed login attempts, Rapid successive actions, Unusual access times, Access from new IP addresses, Bulk data export requests. Triggers admin alerts and optional auto-lockout.

Ø<ß IP Tracking

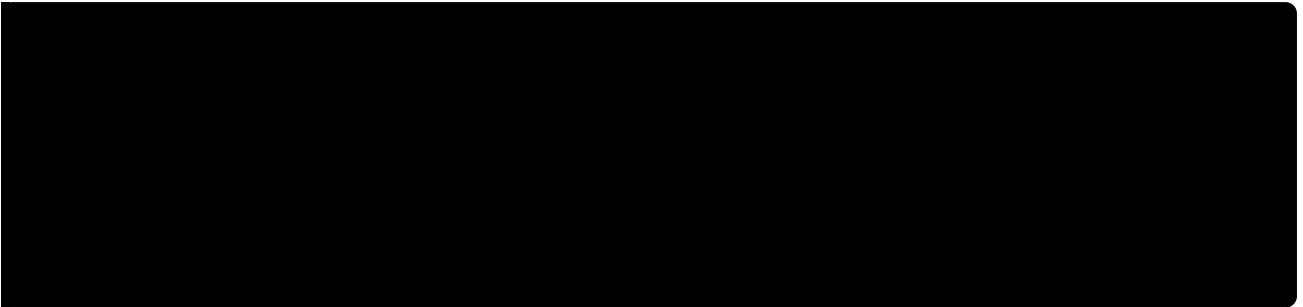
Track and log all IP addresses accessing the admin panel. Whitelist/blacklist management. Geolocation display for admin sessions. Alert on access from unexpected regions. VPN/proxy detection.

Ø=p! Rate Limit Violations

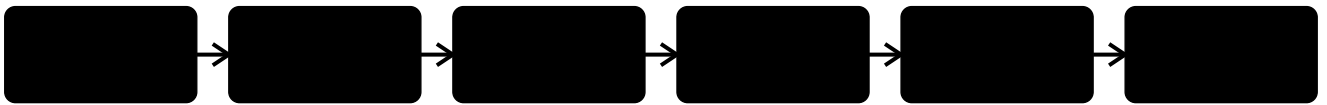
Monitor and log rate limit violations across the platform. Auto-throttle users exceeding limits. Escalating penalties: Warning !' Temporary block (5 min) !' Extended block (1 hour) !' Account suspension. Violation dashboard with patterns.



11. Real-time Update Flow



Complete Real-time Flow



Example 1: Admin Bans User

- Admin clicks "Ban User"
Confirmation dialog ! Admin confirms
- POST /api/admin/users/:id/ban
API route validates admin permission
- Database: user.status = "BANNED"
Prisma updates user record
- Socket.io: "user-banned:{userId}"
Emit to specific user room
- User session invalidated
JWT token blacklisted, session cleared
- User sees "Account Suspended"
Redirect to suspended page immediately

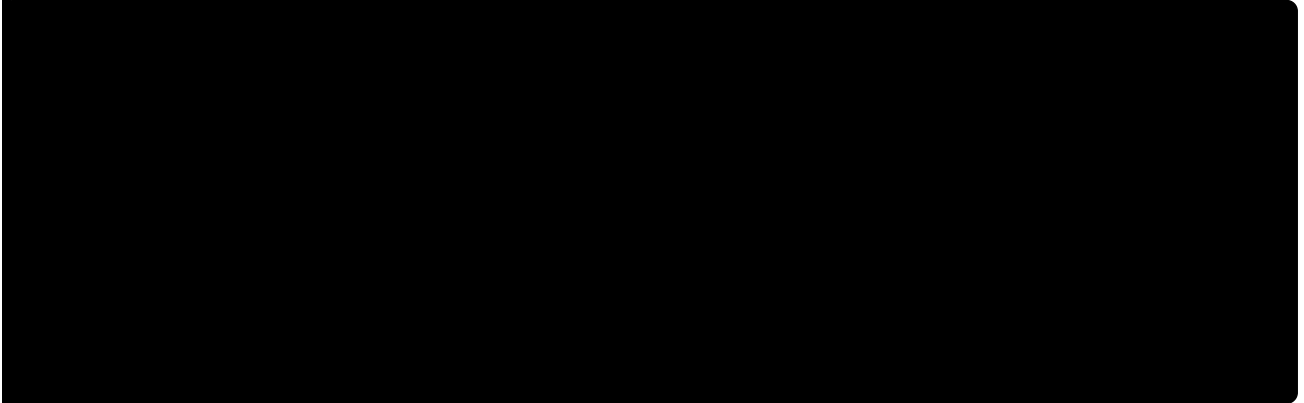
Example 2: Admin Sends Broadcast

- Admin creates notification
Fills title, message, priority
- POST /api/admin/notifications
API saves notification to database
- Socket.io: "admin-notification"
Broadcast to all connected clients
- All users receive instantly
Toast notification + bell icon badge update

Example 3: Admin Toggles Maintenance

- Admin toggles maintenance mode
Sets custom message and estimated recovery time
- POST /api/admin/maintenance
API updates platform settings in database

- Socket.io: "maintenance-mode"
Broadcast to all connected clients with mode & message
- All users see banner immediately
Maintenance banner appears, trading is suspended



All events include: { type, data, timestamp, adminId }